README: DitchExtraction_Toolbox_Feb2026

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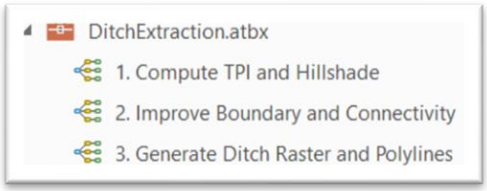
Overview

The **DitchExtraction_Toolbox_Feb2026** folder contains the ArcGIS Pro toolbox, project file, supporting data, and required workspace components for extracting drainage ditch features from a user-supplied Digital Elevation Model (DEM) layer.
It is recommended that all tools be executed within the provided ArcGIS Pro project to ensure proper path management and avoid broken data references.

Folder Contents

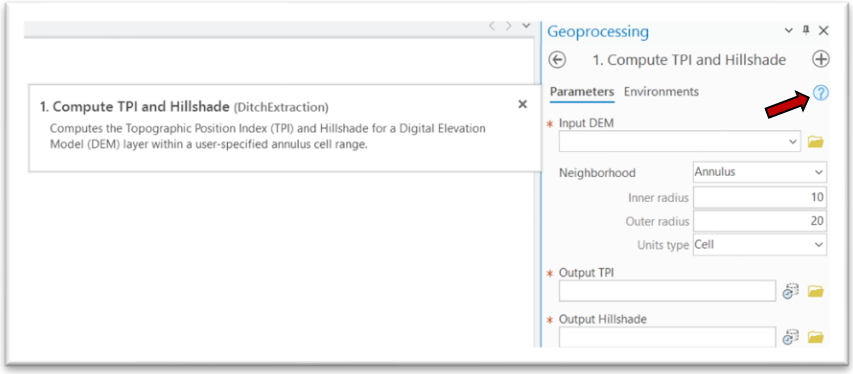
1. **DitchExtraction.atbx**

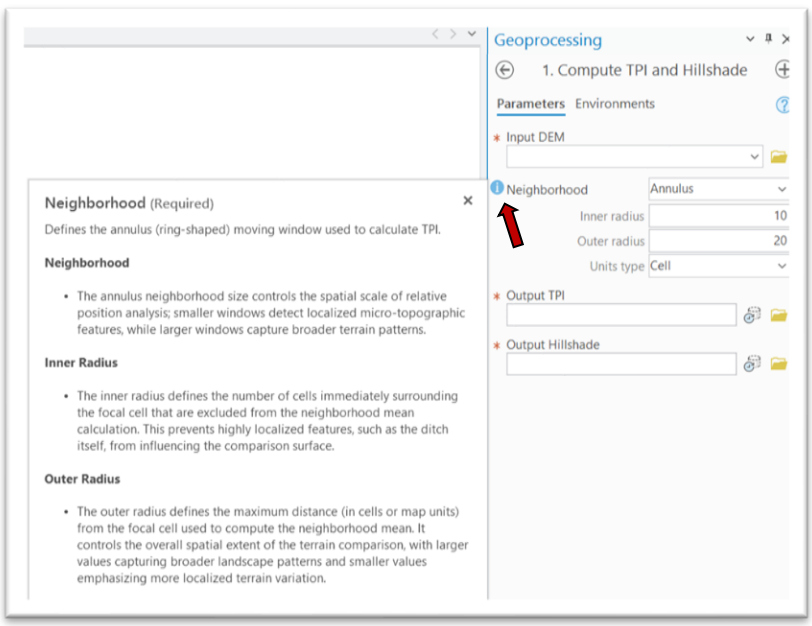


   ArcGIS Pro toolbox containing three tools:
   1. *Compute TPI and Hillshade*
      Generates Topographic Position Index (TPI) and hillshade rasters from the input DEM.
   2. *Improve Boundary and Connectivity*
      Refines thresholded ditch rasters using morphological operations (Boundary Clean, Expand, Shrink, Region Group).
   3. *Generate Ditch Raster and Polylines*
      Filters out small regions (which might be nosiue), generate ditch raster, and converts the raster output to polyline features.
2. **DitchExtraction.aprx**
   - ArcGIS Pro project file configured for this toolbox.
3. **Script Folder**
   a. Contains supporting layer files used by the toolbox for symbology and processing.
4. **Scratch Folder**
   a. Contains Scratch.gdb, which is used as the temporary processing workspace.
5. **Tools.gdb and Tools.atbx**
   a. Default geodatabase and toolbox created automatically by ArcGIS Pro.
   b. Tools.gdb contains a sample DEM covering a small extent within the Hofmann Forest area for testing purposes. The DEM is derived from the USGS 3D Elevation Program (3DEP) Hurricane Florence LiDAR dataset.
      i. The DEM has a spatial resolution of 0.5 meters and in UTM Zone 18N projection.
6. **README**

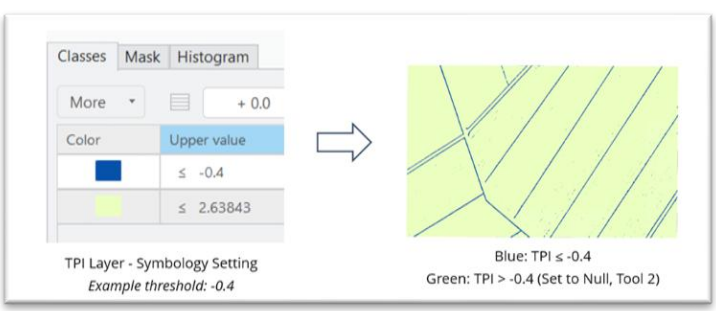
Instructions for DitchExtraction.atbx

- This toolbox consists of three tools designed to extract drainage ditch features from a DEM.
- Detailed descriptions of each tool's purpose, parameters, and input/output instructions can be accessed directly within ArcGIS Pro by hovering over a parameter or clicking the help or information (ⓘ) icon in the tool dialog.

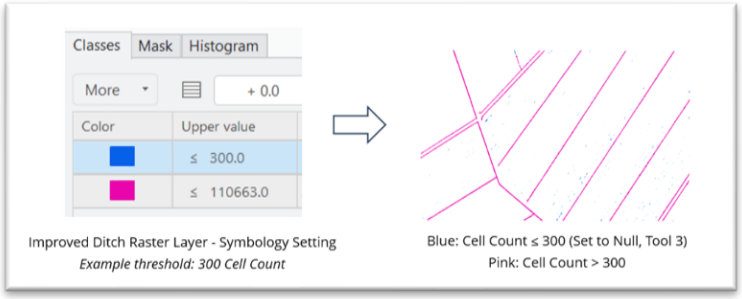




- ***Overall Workflow Logic***
  DEM → Compute TPI *(via Tool 1)*
  → Decide TPI threshold *(via Symbology)*



  → Set TPI threshold and refine raster *(via Tool 2)*
  → Decide minimum region size *(via Symbology)*



→ Set region size and generate final ditch raster and polyline outputs *(Tool 3)*

→ Manually clean the ditch line output if needed

Notes on Output Quality

The quality and accuracy of the ditch products generated by this toolbox depend on several key factors:

- DEM Resolution
  - Higher-resolution DEMs generally produce more accurate and detailed ditch extraction results. Coarse-resolution DEMs may fail to capture narrow or shallow ditches.
- Landscape Characteristics
  - Terrain conditions such as actual ditch depth, surrounding relief, vegetation effects, and tillage can significantly influence detection performance.
- TPI Neighborhood Size (Tool 1)
  - The selected inner and outer radius values control the spatial scale of analysis. Improper neighborhood sizing may under-detect or over-generalize ditch features.
- TPI Threshold Selection (Tool 2)
  - The chosen TPI threshold directly determines which depressions are classified as potential ditch. Threshold selection should be based on visual inspection of the TPI raster.
- Minimum Region Size Threshold (Tool 3)
  - The selected minimum cell count determines how aggressively small regions are removed. Overly high values may remove valid short ditches, while low values may retain noise.

Important

Threshold values and neighborhood parameters are site-specific and should be calibrated for each study area using visual inspection and domain knowledge.

If you have any questions or encounter issues while using this toolbox, please feel free to contact me at emily.yang@duke.edu.